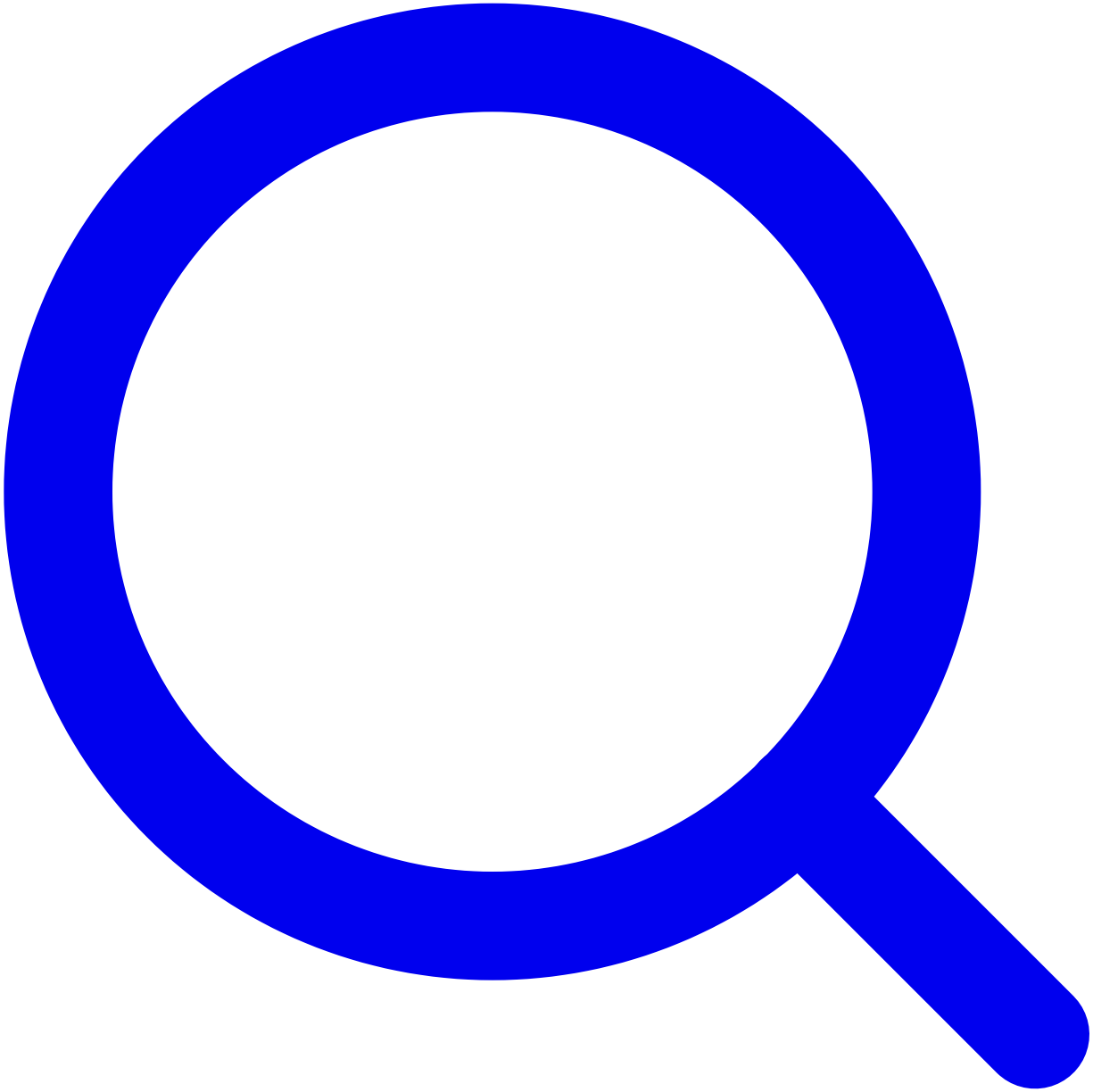


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Astrophysics > Cosmology and Nongalactic Astrophysics

arXiv:2505.06654 (astro-ph)

This paper has been withdrawn by Moustafa Radwan

[Submitted on 10 May 2025 (v1), last revised 3 Apr 2026 (this version, v7)]

## A $G_2$ -Holonomy Model for Late-Time Cosmic Acceleration in M-theory: Alleviating the Hubble Tension through Geometric Vacuum Energy

[Moustafa Amin M. Radwan](#)

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A framework is developed within an eleven-dimensional M-theory scenario where dynamical geometric moduli, originating from a  $G_2$ -holonomy compactification, generate an evolving cosmological term,  $\Lambda(z)$ . This "Geometric Vacuum Energy" (GVE) is shown to be consistent across both cosmological and astrophysical scales. We demonstrate that a natural attractor solution alleviates the Hubble tension, yielding an inferred  $H_0 \approx 69.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$  and a suppressed structure growth parameter  $S_8 \approx 0.67$ , while remaining in

excellent agreement with cosmic chronometer data ( $\chi^2/N \approx 0.44$ ). Its predictions in the strong-gravity regime profoundly strengthen the framework's self-consistency. We show that this model supports stable, hairy black hole solutions whose existence conditions are congruent with the cosmological attractor ( $\Phi_{\text{horizon}} \leftrightarrow \Phi_{\text{cosmology}}$ ). Furthermore, these objects exhibit a rich, falsifiable phenomenology, including (i) the definitive breaking of isospectrality in their gravitational-wave quasi-normal modes, (ii) a suppressed Integrated Sachs-Wolfe effect, and (iii) unique thermodynamic and electromagnetic accretion signatures. These interconnected findings highlight a pathway where a single, theoretically-motivated framework can naturally reconcile multiple observational puzzles.

Comments: *This version is withdrawn due to a fundamental revision of the geometric moduli stabilization parameters. The author has developed a more rigorous first-principles derivation from G<sub>2</sub>-holonomy manifolds that renders the phenomenological assumptions in this early version obsolete*

Subjects: **Cosmology and Nongalactic Astrophysics (astro-ph.CO)**

Cite as: [arXiv:2505.06654](https://arxiv.org/abs/2505.06654) [astro-ph.CO]

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[v6] Wed, 13 Aug 2025 01:24:30 UTC (78 KB)

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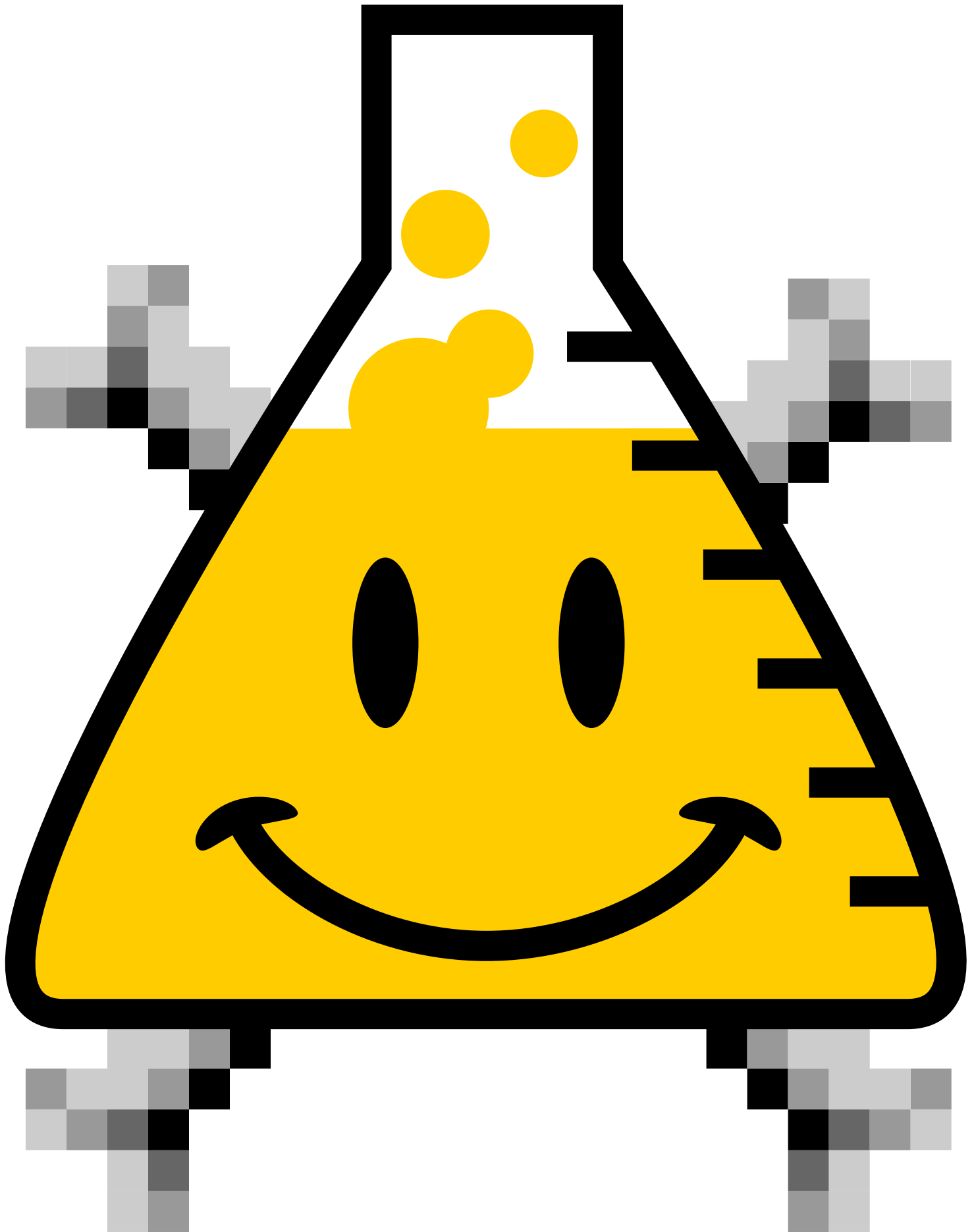
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